



### Wannacry-like ransomware

Source Code for SLocker android ransomware that mimics WannaCry leaked online. <http://digit.in/morecrry>



### Sweden data leak

Personal data of millions of transporters in Sweden, along with the nation's military secrets, have been exposed. <http://digit.in/swedeleak>

protein, fat and fibre. Rumpold and Schlüter (2013) compiled nutrient compositions for 236 edible insects, based on dry matter. Even though this data might have been varied, according to another study (<http://digit.in/mmm-Bugs>), many edible insects provide satisfactory amounts of energy and protein, meet amino acid requirements for humans, are high in monounsaturated and/or polyunsaturated fatty acids, and are rich in micronutrients such as copper, iron, magnesium, manganese, phosphorous, selenium and zinc, as well as riboflavin, pantothenic acid, biotin and, in some cases, folic acid. They're healthy eating!

Many of us want calorie counts instead, and even there, insects usually score above many other meat sources. For instance, according to an FAO 2012 study, an australian green ant, when consumed raw, contributes about 1,272 kcal/100g of fresh weight. Compared to that, the same value for whole chicken is only at 177 kcal/100g of edible portions. According to the same study, protein content of insects, on an average, is at par with that of cattle and fish.

Apart from this, there are specific nutrients available in insects – in quantities significant enough to affect their preexisting consumption around the world. For instance, certain species of caterpillar are rich in lysine (an amino acid used in the biosynthesis of protein). In the Democratic Republic of Congo, they are used to supplement the poor lysine content in their staple diet. Palm weevil larvae play the same role as tubers in the diet of the people of Papua New Guinea. In fact, countries where maize is a staple, there are high number of lysine deficiency cases that can easily be avoided by introducing termite species like *Macrotermes bellicosus* in the diet. Numerous such examples exist around the world.

### WHY SHOULD YOU SWITCH?

Just because insects 'can' substitute many parts of your current flora and fauna-based diet might not be convincing enough for you to make the switch.



Recorded number of edible insect species, by country

But it's not just dietary information that should sway you. One of the primary concerns regarding the current cattle and poultry-based cuisines prevalent around the world is the associated carbon footprint and impact on the environment. Comparatively, consuming insects has a number of environmental advantages:

- They have high feed-conversion efficiency (an animal's capacity to convert feed mass into increased body mass, represented as kg of feed per kg of weight gain).
- They can be reared on organic side streams, reducing environmental contamination, while adding value to waste.
- They emit relatively few Greenhouse gases and relatively little ammonia.
- They require significantly less water than cattle rearing.
- They have few animal welfare issues,

although the extent to which insects experience pain is largely unknown.

- They pose a low risk of transmitting zoonotic infections

Overall, entomophagy has a beneficial effect on the planet without depriving us humans of our dietary requirements. And we all know how important that is!

### A PALATE CLEANSER

There isn't a distinct culture/region in India that has a preference for entomophagy. A large part of today's diet, especially in urban areas, has been influenced by the west. And in the western countries, insects are considered plain disgusting. This westernization of the diet is also affecting areas like Africa and south-east Asia where there is a prevalent practice of entomophagy. This is a harmful trend that needs to be dealt with. Companies and individuals interested in insect farming need to be encouraged by setting up proper infrastructure for the practice. Information about edible insect species and their availability in regions needs to be disbursed as well. But perhaps most importantly, we, the everyday people who run for their slippers at the sight of a cockroach, need to stop looking at these creatures as "creepy-crawlies" that bite us, and learn to bite back, with a side of ketchup, of course! 🍷



Common insect food stall in Bangkok, Thailand. Bon Appetit!

Source: Centre for Geo Information, Wageningen University, based on data compiled by Jongema, 2012.